

Work Experience

Cloudflare	Systems Engineer (DDoS Protection & WAN)	2025 Jan — Present
• High-Performance Networking: Increased single-machine IPsec throughput from 400 Mbps to 5.5 Gbps by writing an eBPF/XDP load balancer in C and Rust that decrypts the initial IPsec block to distribute traffic across all CPU cores. • Post-Quantum Security: Enabled ML-KEM cryptographic key exchange over MTU-constrained paths by implementing IKE IP fragmentation support within the IPsec key service, successfully transmitting encryption payloads exceeding 1024 bytes. • Automated BGP Synchronization: Achieved 100% prefix state consistency between the DDoS configuration service and the BGP Address Management Service by implementing a Go-based event-driven notification listener, backed by a periodic polling mechanism to capture UI-driven configuration changes that bypassed the notification pipeline. • Customer Observability: Reduced monthly L3 support tickets by 60% based on support ticket volume tracking by adding a Rust logger to the IKE service to classify and forward IPsec negotiation logs to an existing telemetry pipeline for customer self-diagnosis. • Engineering Velocity: Reduced CI/CD execution time by 72% (from 18 minutes to 5 minutes) for 120+ DDoS configuration service integration tests by utilizing Postgres templating to parallelize sequential test execution.		
AppLogic Networks	Software Engineer (Simulation Infrastructure)	2020 Sep — 2023 Jul
• Distributed Systems Scaling: Increased control-plane throughput for a 5G simulator from 50,000 to 200,000 messages per second by re-architecting it into a master-slave system with automatic user-partitioned load distribution. • Protocol Engineering: Unblocked 4G/5G cross-generation handover test suites by implementing 3GPP-compliant 5G protocol extensions within an existing C++ 4G core simulator. • Traffic Modeling: Improved hardware capacity planning accuracy from 92% to 98% by building a Python tool that programmatically adjusts packet capture (PCAP) ratios to replicate internet-scale customer traffic profiles. • API Adaptation: Accelerated simulation mode migration by developing a configuration adapter layer to dynamically map test parameters, bypassing an estimated 80+ engineering hours of manual test regression refactoring across 320 stateless test suites.		

Education

Carnegie Mellon University	Master of Software Engineering (GPA: 3.92/4)	2024
• Industry Capstone Project (Team Lead): Reduced point-of-sale data integration latency from 24 hours to 5 minutes at a throughput of 500 transactions per second across 838 retail locations by designing and deploying a Kafka event-streaming pipeline.		
BMS College of Engineering	BE in Electronics & Communication (GPA: 9.04/10)	2020

Technologies and Languages

- **Languages:** C, C++, Rust, Go, Python, SQL
- **Kernel & Networking:** eBPF, XDP, BGP, Network Interconnects, Tunnels (IPsec, GRE), IKE/IP Fragmentation, L3/L4 Routing
- **Infrastructure & Data:** Kafka, Postgres